

Software Defect Prediction Technology Based on Fuzzy Support Vector Machine

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Abstract—Software defect prediction(DP) is one of the most active research areas in software engineering. DP models can provide error-prone source code components or changes, etc., so that quality assurance teams can deal with error-prone source code with more effort, effectively allocating limited resources to verify software products. As software projects grow in size, DP techniques will play a vital role in the work of developers, helping them to develop more reliable software products, resulting in faster time to market. In this paper, fuzzy support vector machine (FSVM) is introduced to build a software DP model, which is trained on the data set by the FSVM classification module. In this paper, by comparing the mean square error of the training data of SVM and FSVM, in the subset of low-defect modules, the training result of FSVM is not as good as that of SVM, that is, the prediction accuracy of SVM is higher; while in the subset of modules with high number of defects, the training result of FSVM is better than that of SVM, that is, the prediction accuracy of FSVM is higher.

Keywords—software defect prediction, fuzzy support vector machine, high defect number module, prediction accuracy.

I. INTRODUCTION

People's increasing reliance on computers increases the likelihood that computer software defects will trigger a crisis. The increase in software scale and software complexity makes the previous software development methods more and more difficult. In order to solve the resulting software crisis, software development methods are gradually being systematic and engineered. However, in the software development process, due to the limitation of resources and developer experience, the existence of software defects is inevitable. Therefore, it is necessary to study the software DP technology, effectively guide the software design process optimization decision, improve the software quality status, and reduce the unnecessary resource investment of developers and the loss of software users as much as possible.

Over the years, predicting the modules that may have defects in software has been paid much attention as an important research direction, and many scholars have also conducted in-depth discussions on DP technology. For example, some scholars have applied K-means and Neural-Gas clustering methods to software module defect classification and prediction. First, they use cluster analysis to classify software modules into two categories according to software measurement data, and then the researchers calibrate the clustering results. There are two types of high defect rate and low defect rate. Cluster analysis is an unsupervised learning method. In the study, the factors that affect the error rate of type I and II are not analyzed, and there is no research on the parameters that can balance the two types of error rates [1]. Some studies have proposed the use of AP (Affinity Propagation) clustering algorithm to predict software defects. The focus of this research is to combine AP algorithm and

genetic algorithm to analyze software metric data, and select the optimal metric subset to make the clustering result within the class I error. A better type II error rate is obtained with a small increase in the rate [2-3]. Comprehensive analysis shows that there are many methods applied to software DP, and further research is needed to improve the performance of DP technology.

This paper first expounds the concept of software defects, builds a DP model based on FSVM for software defects, then analyzes the functions of the four modules in the model framework in detail, and proposes the implementation method of the DP system. Finally, the performance comparison experiment of the DP model is carried out, and the prediction accuracy of the two models is analyzed by comparing the mean variance of SVM and FSVM.

II. SOFTWARE DP BASED ON FSVM

A. The Concept of Software Defects

In the process of software operation, due to the deviation of software documents, data or programs, the software may malfunction or lack functions in some specific environments. If no measures are taken to improve and repair these software defects, software failures will occur [4]. Software DP models are used to detect error-prone modules in an automated manner before manual testing. The project testing team can then allocate time and budget more efficiently based on the results of the model, and the DP module requires more attention than the non-defect prediction module [5-6].

Software DP measurement is a key factor in software quality management. It can not only improve the software process and help project control, but also be used to analyze the quality and performance of software. It has become a powerful tool for software quality assurance. It is indispensable in building statistical forecasting models [7-8].

B. Error-Prone Module of FSVM Classification and Prediction Software

Some studies have applied the fuzzy relation function to assign values of each sample point in the support vector regression experiment, so that the contribution of different sample points in the regression optimization problem can be distinguished [9]. There is a data set as follows: $(x_1, y_1, r_1) \dots (x_i, y_i, r_i) \dots (x_n, y_n, r_n)$, where $0 < r_i < 1$ is the value of the fuzzy relation function corresponding to each sample point, also known as the fuzzy relationship degree, x_i is the input vector, y_i is the target value.

According to the different cost of software DP type I and II errors, the concept of risk control is introduced into the fuzzy support vector machine, and the risk is controlled by

adjusting the error penalty parameter C of the two types of errors [10]. Assuming that the first k samples in the training samples belong to one class, and the remaining $1-k+1$ samples belong to another class, after introducing the concept of risk control, the optimization problem of using FSVM to solve the software module defect classification problem becomes:

$$\min_{w,d,r} \frac{1}{2} w^T w + C_1 \sum_{i=1}^k r_i + C_2 \sum_{i=k+1}^n r_i \quad (1)$$

The constraints are:

$$\begin{aligned} y_i(w^T x_i + d) &\geq 1 - r_i \\ r_i &\geq 0, i = 1, \dots, n \end{aligned} \quad (2)$$

where d is a constant, k and n are sample sizes, and C_1 and C_2 are penalty parameters.

III. SOFTWARE DP MODEL BASED ON FSVM

A. Mining Framework Applied to Software DP

As one of the effective ways to improve software quality, software DP has always been a research hotspot to improve software quality. After long-term research by a large number of scholars, a typical prediction framework has been formed, which is also called the FSVM model, which is a commonly

used model in software DP. The general process of FSVM is to first obtain historical data, then preprocess the data, then divide the data into training data sets and test data sets to train the learner, and finally use the trained learner to perform defects on the new data set. forecast [11].

The FSVM model is mainly divided into two stages: the first stage is the learning stage, which divides the historical data into two parts: training and testing, and the main purpose is to determine the learning plan with better prediction effect. The second stage is the prediction stage, which uses the previous best-performing scheme to make predictions on the new dataset. Evaluation methods can be used in both stages to ensure unbiased results [12].

In software DP, the data used for software DP may contain many features. Generally speaking, the greater the number of features used to construct a defect detection classifier, the better the classification effect. But this is not the case in practical applications. Using all the features to train the classifier has a huge time cost and cannot achieve a good classification effect.

B. Model Framework

The overall framework of the software defect model based on FSVM is shown in Figure 1. The model mainly includes four modules, namely historical data processing module, data preprocessing module, FSVM classification module and software data module to be predicted.

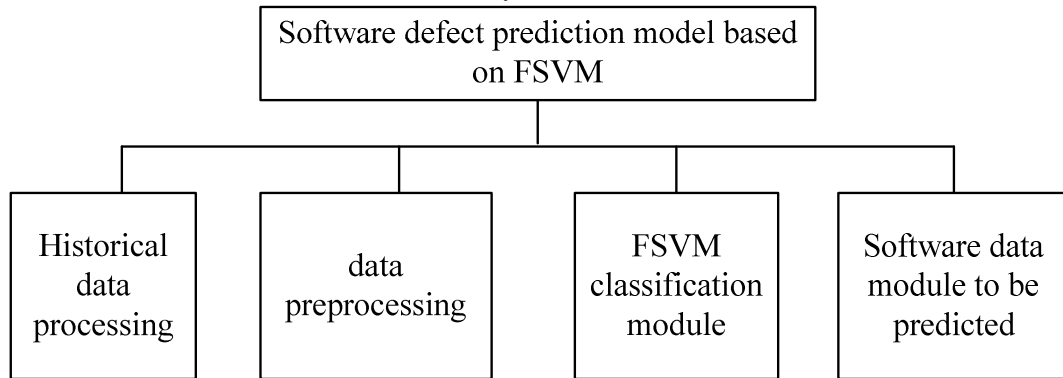


Fig. 1. FSVM-based software defect model

1) The historical data processing module mainly deals with the previous software measurement meta-attribute values and defect information, and the data preprocessing functions include de-noising, de-duplicated data, deletion of erroneous data.

2) Data preprocessing can deal with problems such as noise values and missing values in the data set. Data preprocessing is also a hot research topic in the field of machine learning, and there are different processing methods for different types of data.

Missing value processing: In order to unify the data and facilitate data processing, specific processing is performed on missing values. The simplest processing method is to automatically fill in zeros or ignore them, but it will also interfere with the training model of the data set. The second is to fill in and complete manually, but the problem is that when the data set is too large, this method is difficult to implement. There is also an average value of the sample, which will increase the calculation amount of the model, but the effect is better than directly assigning zero or ignoring it.

Noise value processing: The noise value mainly refers to that the value of some features in the data set is not within the specified range, and the noise value will also affect the accuracy of the prediction model. For the processing of noise values, in addition to the processing of missing values, processing such as regression algorithms can also be used, which can make the data complete and unified.

3) The fuzzy support vector machine classification module uses the training set obtained by the historical data processing module to train the FSVM. After the training is completed, use the test set obtained by the historical data processing module to test the performance of the FSVM, and compare it with the actual value to obtain the prediction accuracy data. In addition, another core function of this module is to obtain the optimal software metric attribute subset and support vector machine parameters after the iterative optimization process, and then the software data module to be predicted provides this module with the metric attribute subset to be predicted. The measurement metadata

of the software under test is obtained, so that the optimized support vector machine predicts whether each module of the software under test has defects.

4) The data module of the software to be tested mainly provides the specific metric value of the software to be tested according to the subset of software metric attributes that is most helpful for DP finally obtained by the genetic optimization module, and inputs it to the fuzzy support vector machine. The classification module is used to predict whether the software module has defects.

C. Implementation of Software DP System

The tools used to implement the DP system are Visual Studio and SQL server. The main functional functions and module interfaces are written in C# language. various data generated.

Before performing DP, you need to import the training set and test set. After the system prompts that the import is successful, select the corresponding preprocessing method, feature selection method and classification learning algorithm. After selecting OK, click Start prediction. The system automatically starts prediction and saves the results. In the prediction results table of the database, click View Result Analysis to see the detailed prediction results processed by a series of methods.

IV. SOFTWARE DP PERFORMANCE ANALYSIS BASED ON SVM AND FSVM

Table I shows the mean square error results of regression training using the entire software metric set as the dataset. The training mean squares of SVM on the MIS and RSDIMU datasets are 42.53 and 5.67, compared to 58.26 and 12.34 for FSVM on MIS and RSDIMU, and the mean squared results of the entire training set show that SVM is better than FSVM.

TABLE I. TRAINING MEAN SQUARE ERROR

	MIS	RSDIMU
SVM	42.53	5.67
FSVM	58.26	12.34

When the fuzzy relationship function is proposed, the fuzzy relationship degree of most modules with fewer defects is reduced, which may cause the prediction accuracy of modules with fewer defects to decrease, but the prediction

accuracy of modules with more defects will definitely be improved. Therefore, we further analyze the training results according to the distinction between low-defect-count modules and high-defect-count modules.

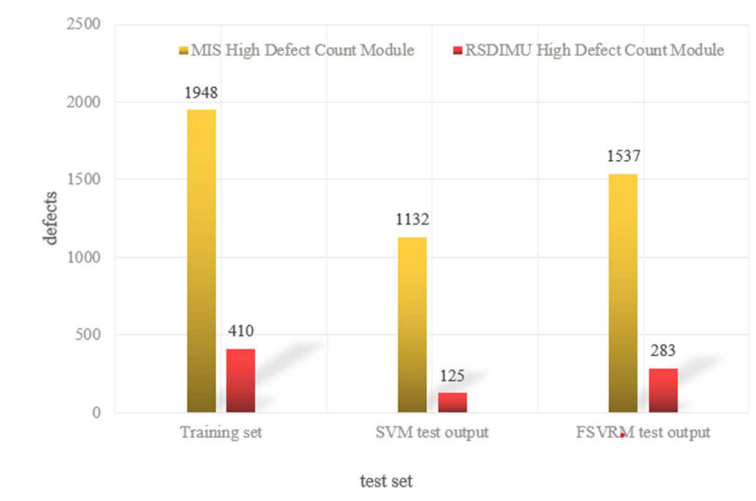


Fig. 2. Total Defect Count for High Defect Count Modules

Figure 2 shows the total defect count for high defect count modules. The sums of module defects marked as high defects in the MIS and RSDIMU training sets are 1948 and 410, respectively; the corresponding results of testing the two training sets by SVM are 1132 and 125, while the

corresponding results of FSVM are 1537 and 283. It can be seen that the number of defects in the test output of FSVM on the training set is much more than that of SVM and is closer to the true number of defects in the module.

TABLE II. MEAN SQUARE ERROR FOR SUBSET OF DEFECT COUNT MODULES

	SVM	FSVM
MIS Low Defect Count Module	19.65	47.33
MIS High Defect Count Module	328.94	156.41
RSDIMU Low Defect Count Module	4.02	9.86
RSDIMU High Defect Count Module	16.18	2.72

Table II is the mean square error data for two subsets of the low-defect-count module and the high-defect-count module. First of all, the mean square error of SVM in the low-

defect module subset of the two datasets is 19.65 and 4.02, respectively, and the mean square error of the corresponding FSVM is 47.33 and 9.86, respectively, indicating that the

training results of FSVM in the low-defect module are not as good as that of SVM. The predictions made before the experiment are verified, that is, the prediction accuracy of modules with fewer defects decreases. Then observe that the mean square deviations of SVM in the subset of high-defect modules of the two datasets are 328.94 and 16.18, respectively, and the mean squares of the corresponding FSVMs are 156.41 and 2.72, respectively, indicating that in the subset of high-defect modules, the training results of FSVM are better than those of SVM. It is much better, which verifies another speculation before the experiment, that is, the prediction accuracy of modules with more defects will definitely be improved.

V. CONCLUSION

This paper proposes a software DP model based on FSVM, which can find defects in the software development process early, locate the defect location, greatly reduce the time for technicians to modify defects, improve development efficiency, and optimize software quality. By comparing the prediction performance of SVM and FSVM, the article shows that although the prediction effect of SVM is better on the entire data set, the prediction effect of SVM and FSVM is different for the high-defect module data set and the low-defect module data set.

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